Nancy Singh

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Education

Sant Longowal Institute of Engineering and Technology (SLIET) — Punjab, India
B.E. in Computer Science and Engineering – SGPA 8.60 — 2026

Delhi Skill and Entrepreneurship University (DSEU) — New Delhi, India
Diploma in Digital Electronics

Projects

Zeitaku | *TypeScript, TypeScript Compiler API, Playwright, Chrome DevTools Protocol, Vitest* — GitHub

- Built a codemod on the **TypeScript Compiler API** that finds hard-coded `waitForTimeout` sleeps in Playwright tests and splices in network-aware waits while preserving the file's formatting.
- Replaced fixed sleeps with **Chrome DevTools Protocol** request tracking: on throttled Slow 3G, fixed 2 s and 3 s sleeps failed **100%** of runs while Zeitaku failed **0%**; **45% faster** than a safe fixed wait on a normal network.
- Scanned **7 open-source codebases (76k files)** and found **240** hard-coded sleeps; AST classification leaves alone the **44** that guard "should not happen" assertions.
- Added flake diagnosis that names the request still in flight when a test fails; **10 unit tests** (incl. a Windows line-ending regression test) run in **GitHub Actions CI**.

DocMind | *Python, FAISS, sentence-transformers, Tesseract OCR, Llama 3.3 (Groq), Streamlit* — Live Demo | GitHub

- Built and deployed a **RAG** app that answers questions about any PDF, running **OCR only on pages without a text layer** so digital PDFs process in seconds.
- Added a retrieval relevance guard that answers "I don't know" **without calling the LLM** when no chunk matches, and shows sources with similarity scores.
- Diagnosed empty answers on scanned PDFs by logging per-page text length; migrated the prototype from ChromaDB to an in-memory **FAISS** cosine index.

HalluGuard | *Python, PyTorch, Hugging Face*

- Built a hallucination detector for RAG answers; moved from cosine similarity to an **NLI cross-encoder (DeBERTa-v3)** and benchmarked it on the **RAGTruth** dataset.

Privacy Analyzer | *Kotlin, TensorFlow Lite* — GitHub

- Built an Android app that runs an ML model **fully on-device** with TensorFlow Lite.

Research

ML Research Project — IIT Jammu
Under Dr. Shaifu Gupta

- Built a machine learning research project for IIT Jammu under the guidance of Dr. Shaifu Gupta.

Open Source

Documenso (open-source DocuSign alternative) — Issue #3445

- Reported about **26 redundant sleeps** in the E2E suite that precede auto-waiting assertions, found through AST analysis, with a fix that keeps sleeps guarding negative assertions.

Technical Skills

Languages: TypeScript, Python, Kotlin, JavaScript
Testing & Browser: Playwright, Chrome DevTools Protocol, Vitest, GitHub Actions, TypeScript Compiler API (AST)
ML & RAG: PyTorch, Hugging Face, sentence-transformers, FAISS, LangChain, NLI cross-encoders, TensorFlow Lite
Tools: Git, Node.js, Express, Streamlit, Tesseract OCR, Linux

Achievements

- **RedBus Data Decode Hackathon:** Top 500 of 8,000+ teams.
- **NCC "C" Certificate** (National Cadet Corps).